STAT 153 & 248 - Time Series
Lecture Twenty Two

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Our next topic is Vector AutoRegressive Models. So far in this course, we have worked with models for a single time series $y_1, \ldots, y_n$. Often, several time series are observed together and they affect each other. Vector AutoRegressive (VAR) models are the simplest models for such data. They are a direct extension of AR models: each series is regressed on the past values of all the series. We shall introduce VAR models through a physical example: the positions of bodies which are connected to each other by springs. Here the laws of physics lead to a specific VAR model, and ignoring the connections between the bodies (i.e., modelling each series separately) leads to predictions which make no sense.

Before discussing VAR models, let us first recall the physical situation from Lecture 16 which was used to motivate univariate AR models.

1 Recap: Spring-Mass AR Motivating Example from Lecture 16

The physical situation is the following. A body of mass m is attached to a horizontal spring (with spring constant κ), and the spring is itself attached to a fixed wall:

```
wall |--/\/\/\/\--[ body ]
        (kappa)
```

When the spring is stretched or compressed and then released, the body oscillates. Our treatment here is slightly different from that of Lecture 16 in two ways:

1. We shall directly work with discrete observations of the position of the body which are taken Δ time units apart (in Lecture 16, we worked in continuous time, where Newton's law is a differential equation).
2. We shall also introduce a frictional force (in Lecture 16, the body was on a frictionless surface).

Let

$$y_t := \text{position of the body at the } t^{th} \text{ time point} \qquad \text{for } t = 1, 2, \ldots.$$

The gap between successive time points is Δ. The position is measured relative to the rest position of the body (this is the position of the body when the spring is neither stretched nor compressed). So $y_t = 0$ means that the body is at its rest position.

Newton's law states that

$$\text{force at time } t \;=\; \text{mass} \times \text{acceleration at time } t.$$

We calculate the two sides of this equation from the discrete observations as follows. There are two forces acting on the body:

1. **Force due to the spring**: By Hooke's law, the spring pulls the body back towards its rest position with a force which is κ times the displacement of the body from its rest position. Using the most recent position y_{t-1}, this force is $-\kappa y_{t-1}$.
2. **Friction**: Friction slows down the body with a force which is γ times the velocity of the body (here γ is the friction coefficient). Using the most recent velocity $(y_{t-1}-y_{t-2})/\Delta$, this force is $-\gamma(y_{t-1}-y_{t-2})/\Delta$.

The total force at time t is therefore

$$-\kappa y_{t-1} - \gamma\,\frac{y_{t-1}-y_{t-2}}{\Delta}.$$

For the acceleration, note that the velocity of the body between the times $t-1$ and t is $(y_t - y_{t-1})/\Delta$, and its velocity between the times $t-2$ and $t-1$ is $(y_{t-1}-y_{t-2})/\Delta$. The acceleration is the change in the velocity divided by Δ:

$$\frac{(y_t - y_{t-1}) - (y_{t-1}-y_{t-2})}{\Delta^2} = \frac{y_t - 2y_{t-1} + y_{t-2}}{\Delta^2}.$$

Newton's law thus gives

$$-\kappa y_{t-1} - \gamma\,\frac{y_{t-1}-y_{t-2}}{\Delta} = m\,\frac{y_t - 2y_{t-1} + y_{t-2}}{\Delta^2}.$$

Multiplying both sides by Δ^2/m, we get

$$y_t - 2y_{t-1} + y_{t-2} = -\frac{\kappa\Delta^2}{m}y_{t-1} - \frac{\gamma\Delta}{m}\,(y_{t-1}-y_{t-2})$$

which is the same as

$$y_t = \left(2 - \frac{\kappa\Delta^2}{m} - \frac{\gamma\Delta}{m}\right)y_{t-1} - \left(1 - \frac{\gamma\Delta}{m}\right)y_{t-2}.$$

With the notation

$$a := \frac{\kappa\Delta^2}{m} \quad \text{and} \quad g := \frac{\gamma\Delta}{m},$$

this becomes $y_t = (2-a-g)y_{t-1} - (1-g)y_{t-2}$. To construct a model, we add a noise term ϵ_t:

$$y_t = (2-a-g)y_{t-1} - (1-g)y_{t-2} + \epsilon_t \qquad \text{where } \epsilon_t \overset{\text{i.i.d}}{\sim} N(0,\sigma^2). \tag{1}$$

As in Lecture 16, the noise ϵ_t represents the effect of random disturbances to the body (imagine kids randomly throwing stones at the body while it is oscillating). The model (1) is an example of the AR(2) model $y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$: here $\phi_0 = 0$, $\phi_1 = 2-a-g$ and $\phi_2 = -(1-g)$. When there is no friction (i.e., when $\gamma = 0$), we get $\phi_2 = -1$, which is Yule's model from Lecture 16. The only approximation that we made is in replacing derivatives (velocity and acceleration) by differences. This is accurate when Δ is small compared to the period of the oscillations.

2 Two Connected Bodies

Now consider two bodies, each of mass m, which are placed between two walls. Each body is attached to the wall next to it by a spring with spring constant κ, and the two bodies are connected to each other by a third spring with spring constant κ_c:

```
wall |--/\/\/\/\--[ body 1 ]--/\/\/\/\--[ body 2 ]--/\/\/\/\--| wall
       (kappa)                (kappa_c)                (kappa)
```

As before, the positions of the two bodies are recorded at time points which are Δ apart. Let

$$y_t(i) := \text{position of body } i \text{ at the } t^{th} \text{ time point} \qquad \text{for } i = 1, 2 \text{ and } t = 1, 2, \ldots$$

and let

$$y_t := \begin{pmatrix} y_t(1) \\ y_t(2) \end{pmatrix}$$

denote the 2×1 vector of the two positions at time t. Each position is measured as the displacement (towards the right) of the body from its rest position (the rest positions are the positions of the bodies when none of the springs is stretched or compressed). So $y_t(i) = 0$ means that body i is at its rest position.

Let us write down Newton's law for body 1 in the same way as in the last section. The forces acting on body 1 at time t are:

1. The spring on its left is stretched by $y_{t-1}(1)$, so it exerts the force $-\kappa y_{t-1}(1)$.
2. The spring between the two bodies is compressed by $y_{t-1}(1) - y_{t-1}(2)$, so it exerts the force $-\kappa_c \left(y_{t-1}(1) - y_{t-1}(2)\right)$.
3. Friction exerts the force $-\gamma(y_{t-1}(1) - y_{t-2}(1))/\Delta$ (we take the friction coefficient γ to be the same for both the bodies).

Newton's law therefore gives

$$-\kappa y_{t-1}(1) - \kappa_c \left(y_{t-1}(1) - y_{t-1}(2)\right) - \gamma \, \frac{y_{t-1}(1) - y_{t-2}(1)}{\Delta} = m \, \frac{y_t(1) - 2y_{t-1}(1) + y_{t-2}(1)}{\Delta^2}.$$

This is the same as

$$y_t(1) = \left(2 - \frac{\kappa\Delta^2}{m} - \frac{\kappa_c\Delta^2}{m} - \frac{\gamma\Delta}{m}\right) y_{t-1}(1) + \frac{\kappa_c\Delta^2}{m} y_{t-1}(2) - \left(1 - \frac{\gamma\Delta}{m}\right) y_{t-2}(1).$$

Note that the right hand side involves $y_{t-1}(2)$: the position of body 1 at time t depends not only on its own past positions, but also on the past position of body 2. The equation for body 2 is derived in the same way (the forces acting on body 2 are $\kappa_c \left(y_{t-1}(1) - y_{t-1}(2)\right)$ from the spring between the two bodies, $-\kappa y_{t-1}(2)$ from the spring on its right, and friction). With the notation

$$a := \frac{\kappa\Delta^2}{m} \qquad b := \frac{\kappa_c\Delta^2}{m} \qquad g := \frac{\gamma\Delta}{m},$$

the two equations are

$$\begin{aligned} y_t(1) &= (2 - a - b - g) y_{t-1}(1) + b y_{t-1}(2) - (1 - g) y_{t-2}(1) \\ y_t(2) &= (2 - a - b - g) y_{t-1}(2) + b y_{t-1}(1) - (1 - g) y_{t-2}(2). \end{aligned}$$

To construct a model, we add noise terms $\epsilon_t(1)$ and $\epsilon_t(2)$ (they represent random disturbances to the two bodies):

$$\begin{aligned} y_t(1) &= (2-a-b-g)y_{t-1}(1) + by_{t-1}(2) - (1-g)y_{t-2}(1) + \epsilon_t(1) \\ y_t(2) &= (2-a-b-g)y_{t-1}(2) + by_{t-1}(1) - (1-g)y_{t-2}(2) + \epsilon_t(2). \end{aligned} \tag{2}$$

It is convenient to write (2) in matrix notation. Let ϵ_t denote the 2×1 vector with entries $\epsilon_t(1)$ and $\epsilon_t(2)$. Then (2) is the same as

$$y_t = \begin{pmatrix} 2-a-b-g & b \\ b & 2-a-b-g \end{pmatrix} y_{t-1} + \begin{pmatrix} -(1-g) & 0 \\ 0 & -(1-g) \end{pmatrix} y_{t-2} + \epsilon_t.$$

Denoting the two matrices above by A_1 and A_2 respectively, this is

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + \epsilon_t. \tag{3}$$

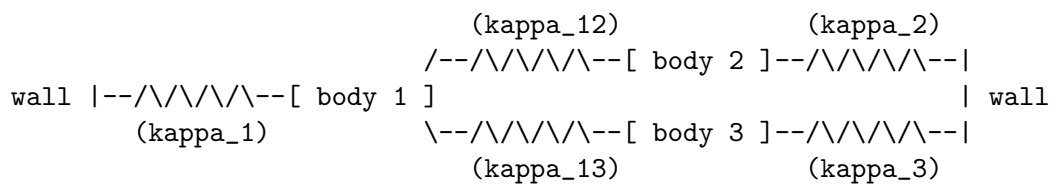
This looks just like the AR(2) model (1), except that the observation y_t at each time is now a vector, and the coefficients A_1 and A_2 are matrices. It is an example of a Vector AutoRegressive (VAR) model. Note that if $b = 0$ (i.e., if there is no spring between the two bodies), then (2) is just two separate AR(2) models of the form (1), one for each body. The connection between the two bodies appears in the off-diagonal entries of the matrix A_1.

3 k Connected Bodies

The same argument works for any number of bodies and for any pattern of connections between them. Consider k bodies which move along a line, and let $y_t(i)$ denote the position of body i at the tth time point (measured as before from its rest position). Suppose that:

1. body i has mass m_i and friction coefficient γ_i,
2. body i is connected to a wall by a spring with spring constant κ_i (we take $\kappa_i = 0$ if body i is not connected to a wall),
3. bodies i and j are connected to each other by a spring with spring constant $\kappa_{ij} = \kappa_{ji}$ (we take $\kappa_{ij} = 0$ if bodies i and j are not connected).

Here is an example with $k = 3$ bodies:



In this example, body 1 is connected to bodies 2 and 3, while bodies 2 and 3 are not connected to each other (so $\kappa_{23} = 0$).

The total force exerted by the springs on body i at time t equals

$$-\kappa_i y_{t-1}(i) - \sum_{j:j\neq i} \kappa_{ij} \left(y_{t-1}(i) - y_{t-1}(j) \right),$$

and it can be checked (by the same argument as in the last section) that the model equations become

$$\begin{aligned} y_t(i) = &\left(2 - \frac{\Delta^2\kappa_i}{m_i} - \sum_{j:j\neq i}\frac{\Delta^2\kappa_{ij}}{m_i} - \frac{\Delta\gamma_i}{m_i}\right) y_{t-1}(i) + \sum_{j:j\neq i}\frac{\Delta^2\kappa_{ij}}{m_i} y_{t-1}(j) \\ &- \left(1 - \frac{\Delta\gamma_i}{m_i}\right) y_{t-2}(i) + \epsilon_t(i) \end{aligned} \tag{4}$$

for $i = 1, \ldots, k$. In matrix notation, this is again the equation (3), where y_t and ϵ_t are now $k \times 1$ vectors, and A_1 and A_2 are $k \times k$ matrices. We shall denote the $(i,j)^{th}$ entry of a matrix A by $A(i,j)$. The entries of A_1 and A_2 can be read off from (4). The matrix A_2 is diagonal with $A_2(i,i) = -(1 - \Delta\gamma_i/m_i)$, the diagonal entries of A_1 are the coefficients of $y_{t-1}(i)$ in (4), and the off-diagonal entries of A_1 are given by

$$A_1(i,j) = \frac{\Delta^2\kappa_{ij}}{m_i} \qquad \text{for } i \neq j. \tag{5}$$

Thus, for $i \neq j$, the entry $A_1(i,j)$ is nonzero exactly when bodies i and j are connected by a spring: the matrix A_1 tells us which bodies are connected to which. For instance, in the example with three bodies above,

$$A_1 = \begin{pmatrix} * & \Delta^2\kappa_{12}/m_1 & \Delta^2\kappa_{13}/m_1 \\ \Delta^2\kappa_{12}/m_2 & * & 0 \\ \Delta^2\kappa_{13}/m_3 & 0 & * \end{pmatrix}$$

where $*$ denotes the diagonal entries (the two zeros are there because bodies 2 and 3 are not connected to each other). Also, because $A_1(i,j) = \Delta^2\kappa_{ij}/m_i$ and $A_1(j,i) = \Delta^2\kappa_{ij}/m_j$, if body i is much heavier than body j, then $A_1(i,j)$ is much smaller than $A_1(j,i)$: the heavy body moves the light body, but the light body hardly moves the heavy body.

4 The VAR Model

The models of the last two sections are of the form (3). If we also include an intercept term c, we get

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \epsilon_t.$$

This is called the Vector AutoRegressive model of order two, or VAR(2). More generally, suppose that we have k time series, and let y_t denote the $k \times 1$ vector consisting of the observations of the k series at time t. The Vector AutoRegressive model of order p (denoted by VAR(p)) is given by

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + \epsilon_t \qquad \text{where } \epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \Sigma). \tag{6}$$

Here c is a $k \times 1$ vector, $A_1, \ldots, A_p$ are $k \times k$ matrices, and Σ is a $k \times k$ covariance matrix. The error vector ϵ_t is assumed to be independent of the past values $y_{t-1}, y_{t-2}, \ldots$. The ith equation in (6) is

$$y_t(i) = c(i) + \sum_{j=1}^{k} A_1(i,j) y_{t-1}(j) + \cdots + \sum_{j=1}^{k} A_p(i,j) y_{t-p}(j) + \epsilon_t(i). \tag{7}$$

Each of the k series at time t is thus a linear combination of the p most recent values of all the k series, plus noise. When $k = 1$, this is the AR(p) model.

For connected bodies, we have $p = 2$ and $c = 0$ (if the positions are not measured from the rest positions, then c is nonzero, just like ϕ_0 in Lecture 16), and the matrices A_1 and A_2 have the special form described in the last section. If the random disturbances to the different bodies are independent, then Σ is a diagonal matrix. In general, the errors of the different series at the same time are allowed to be correlated with each other.

The unknown parameters in (6) are $c, A_1, \ldots, A_p$ and Σ. Each of the k equations has $kp + 1$ coefficients, so there are $k(kp + 1)$ coefficients in total (for example, 55 coefficients when $k = 5$ and $p = 2$). For connected bodies, most of these coefficients are zero, and the others are determined by a few physical constants. But in practice, we usually do not know these constants (we may not even know which bodies are connected to which), and for most datasets there is no physical theory at all. So we treat all the parameters as unknown and estimate them from the data.

5 Parameter Estimation

Suppose that the observed data are $y_1, \ldots, y_n$ (each y_t is a $k \times 1$ vector). Each of the k equations (7) is, by itself, a linear regression model: the response is $y_t(i)$, and the covariates are the p most recent values of all the k series. So the VAR(p) model is fit by k separate regressions, one for each series. For each $i = 1, \ldots, k$, we regress

$$y_t(i) \quad \text{on} \quad 1,\ y_{t-1}(1), \ldots, y_{t-p}(1),\ y_{t-1}(2), \ldots, y_{t-p}(2),\ \ \ldots\ ,\ y_{t-1}(k), \ldots, y_{t-p}(k) \qquad (8)$$

using the observations for $t = p + 1, \ldots, n$. The estimated coefficients in the ith regression are the estimates of $c(i)$ and of the ith rows of the matrices $A_1, \ldots, A_p$. Note that all the k regressions have the same $kp + 1$ covariates (only the response changes from one regression to the next).

It is instructive to compare (8) with the AR(p) model for the ith series alone, which is fit by regressing

$$y_t(i) \quad \text{on} \quad 1,\ y_{t-1}(i), \ldots, y_{t-p}(i).$$

The VAR(p) model thus adds the past values of all the other series as additional covariates in the regression for each series.

Each of the k regressions is fit by least squares in the usual way (exactly as for AR models), and the standard errors of the estimated coefficients are also obtained in the usual way. The covariance matrix Σ of the errors is estimated from the residuals of the k regressions.

In Python, the VAR(p) model can be fit by the function `VAR` in `statsmodels`. This is done via `VAR(data).fit(p)`, where `data` is the $n \times k$ matrix of observations (its tth row is y_t). The results coincide with those obtained by running `OLS` separately for each of the k regressions. As in the case of AR models, the order p can be chosen by AIC or BIC (this is done by the function `select_order`).

6 Prediction

Predictions for future values are obtained in the same way as for AR models. In the VAR(p) model, the predictions $\hat{y}_{n+h}$ of y_{n+h} are obtained by the recursion:

$$\hat{y}_{n+h} = \hat{c} + \hat{A}_1 \hat{y}_{n+h-1} + \hat{A}_2 \hat{y}_{n+h-2} + \cdots + \hat{A}_p \hat{y}_{n+h-p} \qquad \text{for } h = 1, 2, \ldots \qquad (9)$$

which is initialized by $\hat{y}_j = y_j$ for $j = n, n-1, \ldots, n+1-p$.

7 Granger Causality

A basic question about several time series is: which series help in predicting which? Consider the equation (7) for the ith series in the VAR(p) model, and fix another series $j \neq i$. If

$$A_1(i,j) = A_2(i,j) = \cdots = A_p(i,j) = 0, \tag{10}$$

then the past values of series j do not appear in the equation for $y_t(i)$. In this case, once the past values of the other series are used, the past values of series j are of no additional use for predicting $y_t(i)$. If (10) is not true, then we say that series j **Granger-causes** series i (this notion was introduced by Granger in 1969). In other words, series j Granger-causes series i if the past values of series j help in predicting $y_t(i)$, over and above the past values of all the other series (including series i itself).

For connected bodies, we have $p = 2$, the matrix A_2 is diagonal, and $A_1(i,j)$ is given by (5). So body j Granger-causes body i exactly when the two bodies are connected by a spring (for example, if five bodies are arranged in a row, each body Granger-causes its neighbours and no other body). Also, if body i is very heavy, then $A_1(i,j)$ is essentially zero: a heavy body is not Granger-caused by the light bodies which are connected to it (while it does Granger-cause them).

When the matrices $A_1, \ldots, A_p$ are estimated from data, the estimates are never exactly zero. Because Granger causality is about prediction, the most direct way of assessing it is to compare the accuracy of predictions of $y_t(i)$ with and without the past values of series j. Each prediction is obtained by regressing $y_t(i)$ on the p most recent values of the series that are being used. We fit the two regressions using the first part of the data, use them to predict $y_t(i)$ for each t in the remaining part of the data (from the actual values at the times $t-1, \ldots, t-p$), and compare the RMSEs of the two sets of predictions.